Assessment Dark Forest Theory in Multi-Agent Reinforcement Learning

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"But where is everybody?" Enrico Fermi

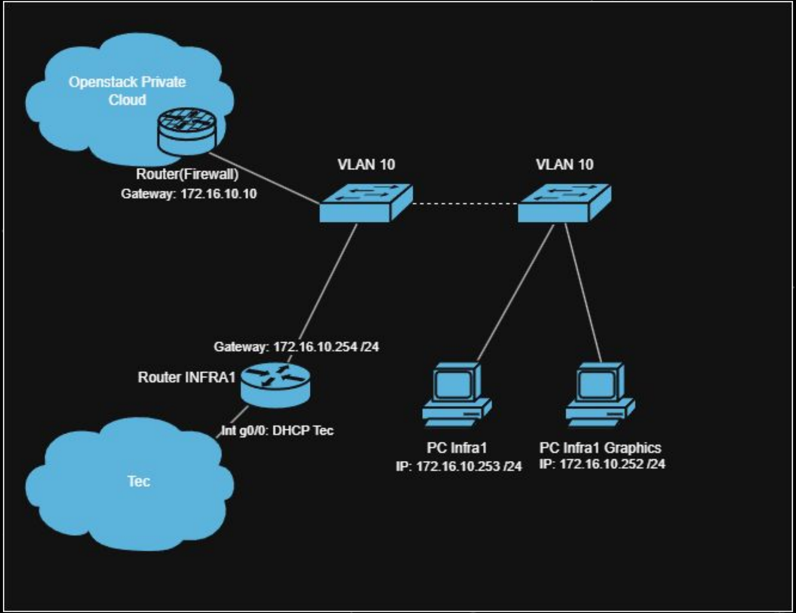
Dark Forest Theory

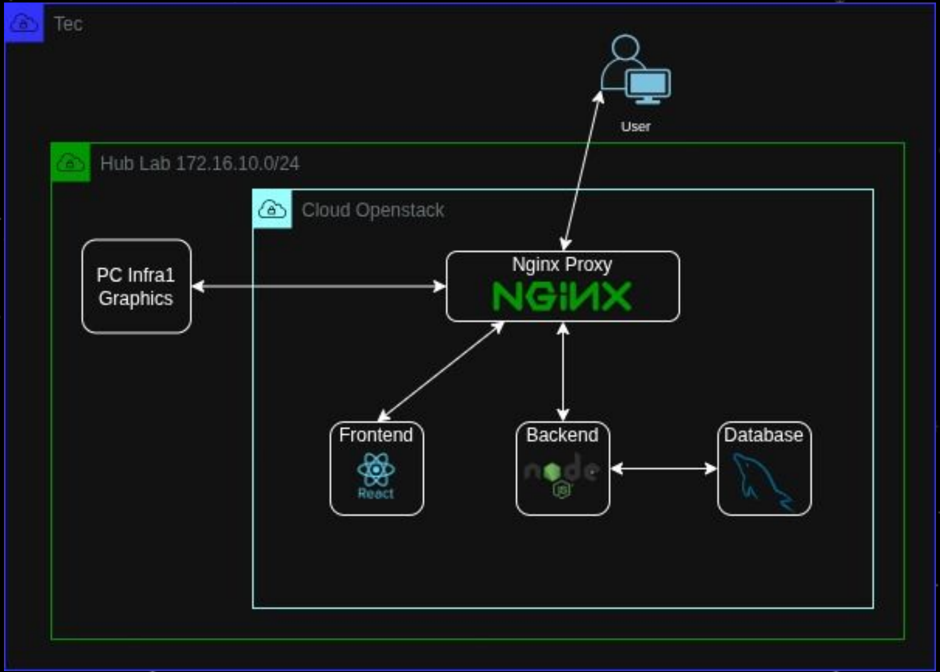
A possible answer to the Fermi paradox:

Perhaps everyone is there, but everyone is **hiding**.

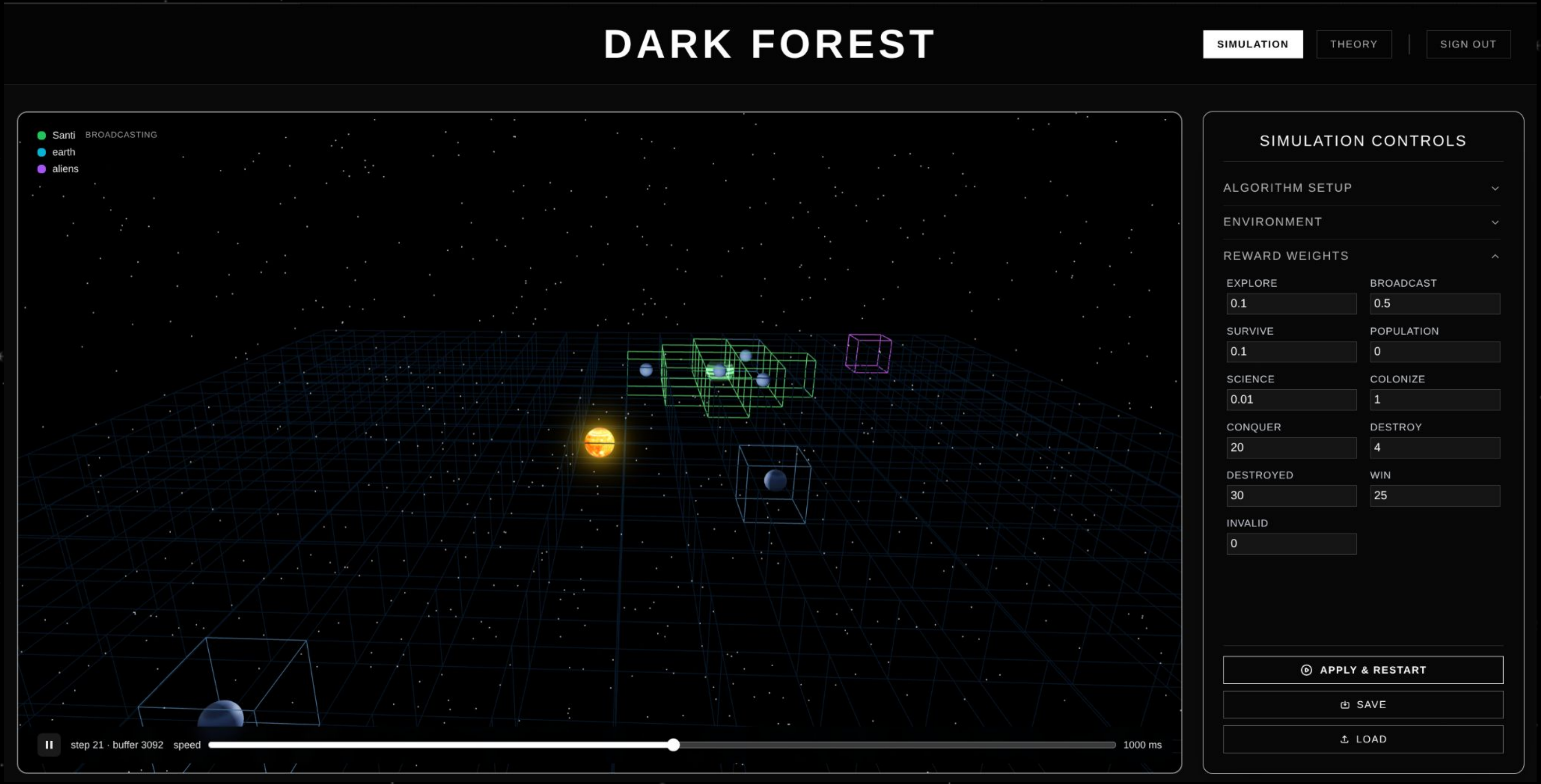
Called so for Liu Cixin's literary series: The Three Body Problem

01
Infraestructura


Openstack Private Cloud
Router(Firewall)
Gateway: 172.16.10.10
VLAN 10
VLAN 10
Gateway: 172.16.10.254 /24
Router INFRA1
Int g0/0: DHCP Tec
Tec
PC Infra1
IP: 172.16.10.253 /24
PC Infra1 Graphics
IP: 172.16.10.252 /24


Tec
User
Hub Lab 172.16.10.0/24
Cloud Openstack
PC Infra1 Graphics
Nginx Proxy
NGINX
Frontend
React
Backend
node
Database

02
Aplicación Web


DARK FOREST
SIMULATION
THEORY
SIGN OUT
Santi BROADCASTING
earth
aliens
step 21 · buffer 3092 speed
1000 ms
SIMULATION CONTROLS
ALGORITHM SETUP
ENVIRONMENT
REWARD WEIGHTS
EXPLORE
0.1
BROADCAST
0.5
SURVIVE
0.1
POPULATION
0
SCIENCE
0.01
COLONIZE
1
CONQUER
20
DESTROY
4
DESTROYED
30
WIN
25
INVALID
0
APPLY & RESTART
SAVE
LOAD

03

Multi Agent Proximal Policy Optimization

Our Hypothesis

Given an appropriate interplanetary environment, Multi-Agent Reinforcement Learning (MARL) could naturally reach a state partially or fully described by the dark forest theory.

What is Proximal Policy Optimization?

Is a deep RL algorithm for improving the performance of models by policies that indicates how an agent has learned to act in the world .

A policy is a strategy or set of rules that the agent follows to select a discrete action based on its environment.

The idea is to discover an **optimal value function** for each possible action that will lead to an optimal policy.

Why Proximal policy optimization?

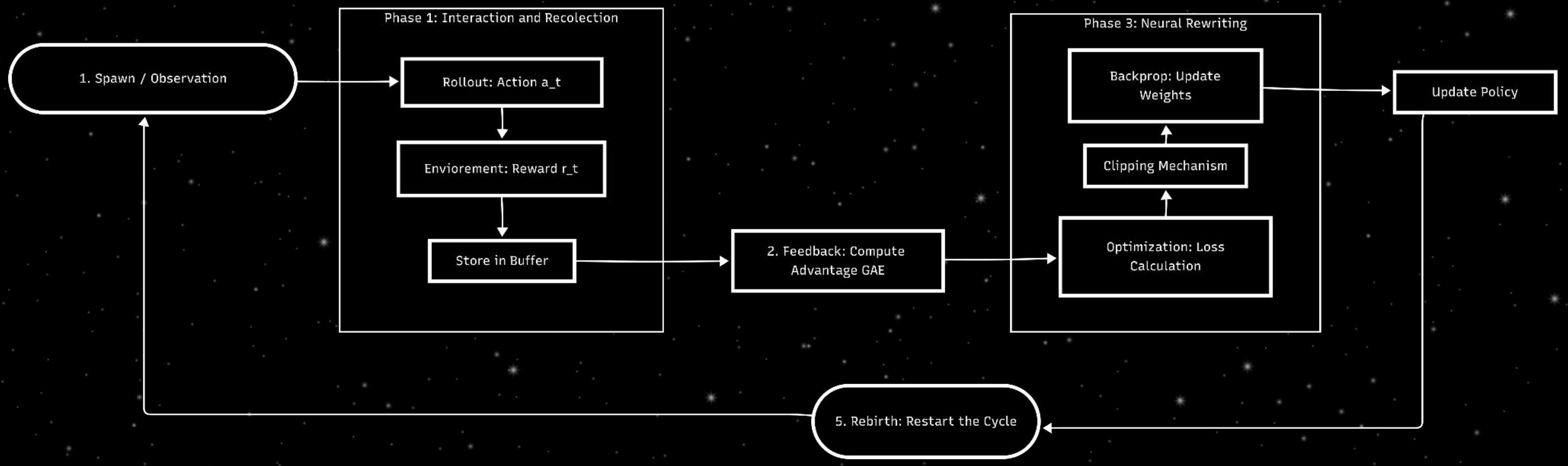
The environment is changing because of the multiple agents: so off policy would get confused by the stale observations.

Sampling Efficiency with Complex Variables

An excellent balance between **exploration and exploitation** through the use of the advantage

Multi Agent support

Pipeline


1. Spawn / Observation
Phase 1: Interaction and Recolection
Rollout: Action a_t
Envioremеnt: Reward r_t
Store in Buffer
2. Feedback: Compute Advantage GAE
Phase 3: Neural Rewriting
Optimization: Loss Calculation
Clipping Mechanism
Backprop: Update Weights
Update Policy
5. Rebirth: Restart the Cycle

Civilizations

13 attributes and 8 actions

Planets

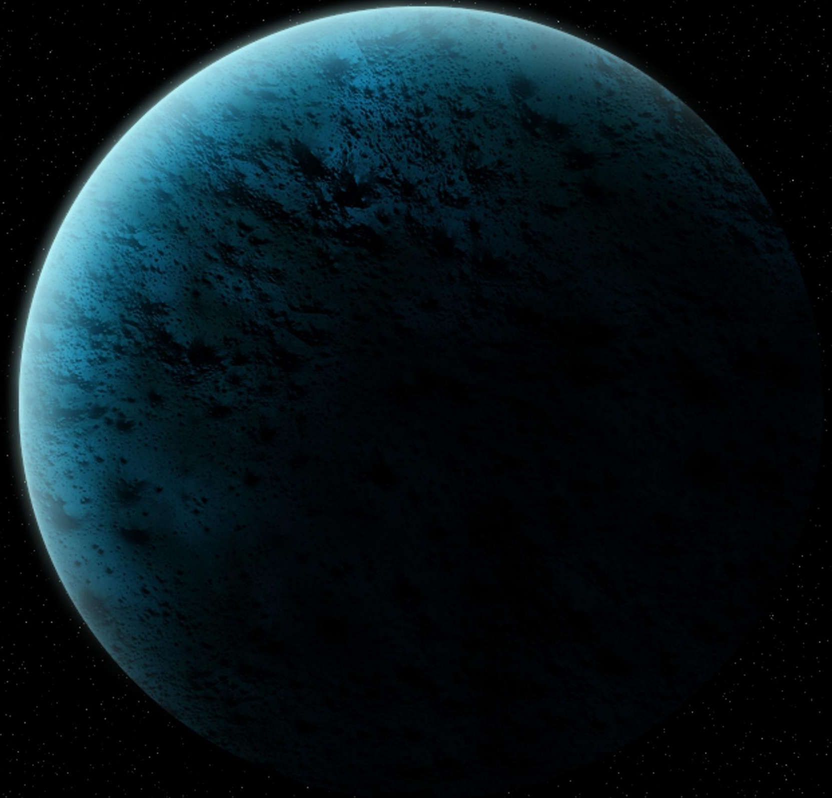
Hold civilizations and resources

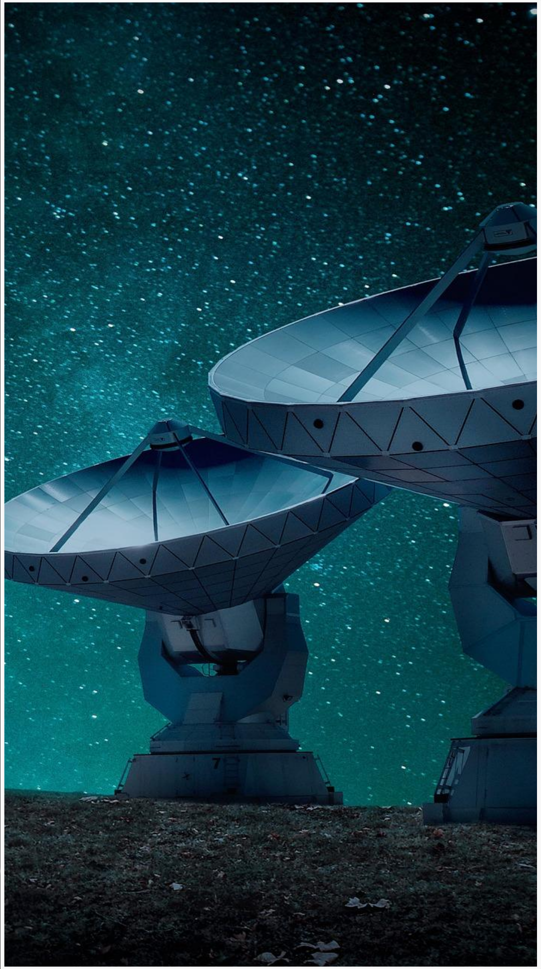
Galaxy

Holds all planets

Civilization Actions and rewards

- Explore
- Survive
- Broadcast position
- Colonize planet
- Destroy Planet
- Earn science points





Civilizations Stopped Broadcasting

Ignoring rewards for broadcasting,
in SOME configurations.
Ignoring population rewards and after 400 iters

Conclusions and future work

Implement additional security controls for the application and quality assurance

Implementation of additional RL algorithms such as SAC and QMIX,

Given customization, application sin more areas such as banking, game theory, cognitive science

Thank you!

References

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